

Security Audit

OnlyUp (Token)

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CAUTION

THIS DOCUMENT IS A SECURITY AND FUNCTIONALITY AUDIT REPORT AND MAY CONTAIN CONFIDENTIAL INFORMATION. THIS INCLUDES IDENTIFIED VULNERABILITIES AND MALICIOUS CODE WHICH COULD BE USED TO COMPROMISE THE PROJECT. THIS DOCUMENT SHOULD ONLY BE FOR INTERNAL USE UNTIL ISSUES ARE RESOLVED. ONCE VULNERABILITIES ARE REMEDIATED, THIS REPORT CAN BE MADE PUBLIC. THE CONTENT OF THIS REPORT IS OWNED BY HASHLOCK PTY LTD FOR USE OF THE CLIENT.

Executive Summary

The OnlyUp team partnered with Hashlock to conduct a security audit of their OnlyUpOnEthereum.sol and OnlyUpGeneric.sol smart contracts. Hashlock manually and proactively reviewed the code in order to ensure the project's team and community that the deployed contracts were secure and functional.

Project Context

OnlyUp introduces a token model where buying pressure drives significant value appreciation. This is achieved through a linear unlock mechanism, which gradually releases tokens in line with market activity, ensuring long-term growth.

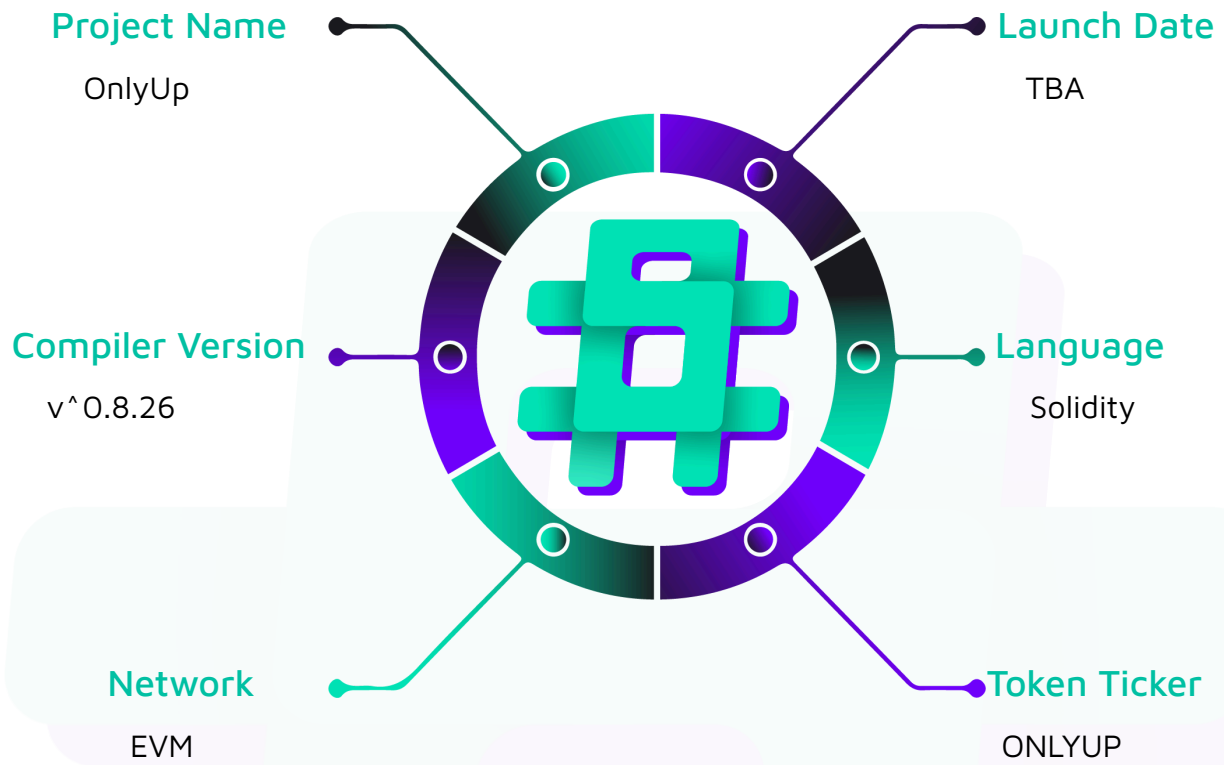
Project Name: OnlyUp

Compiler Version: ^0.8.26

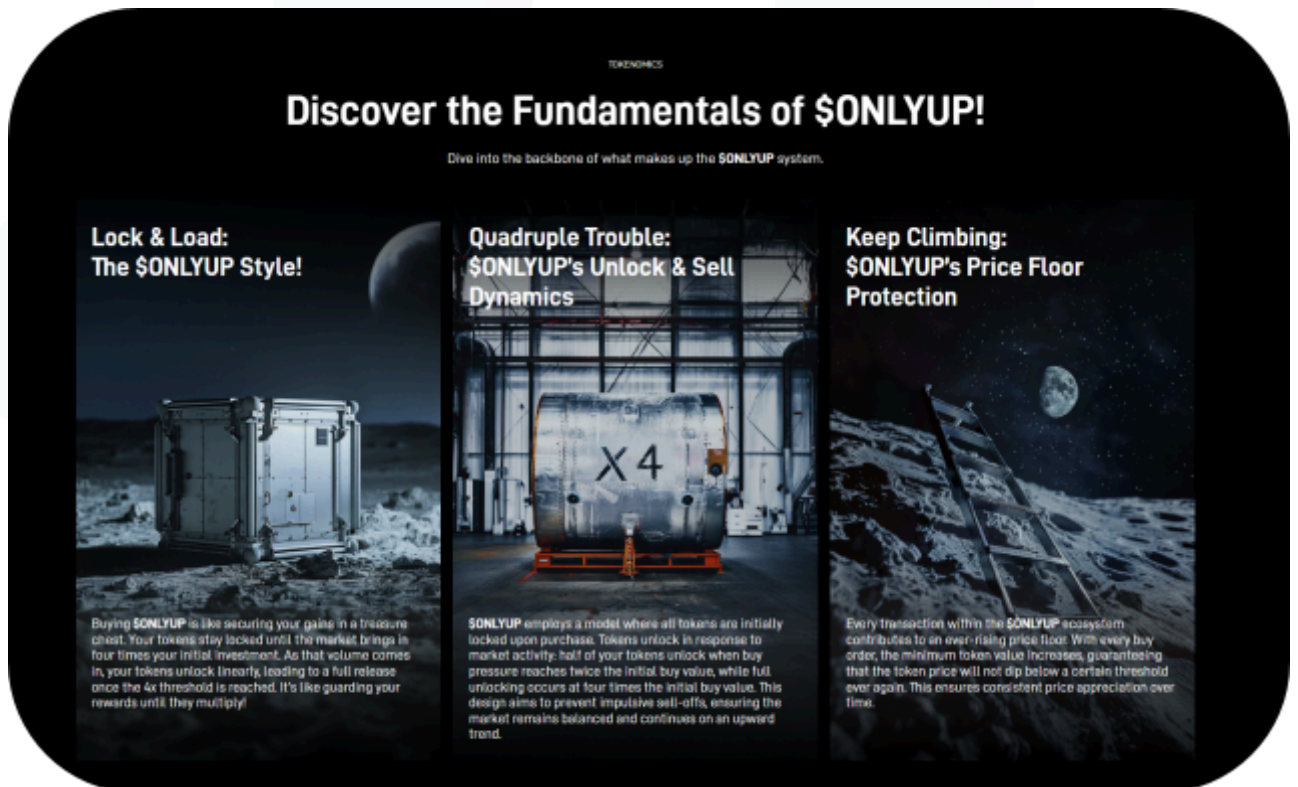
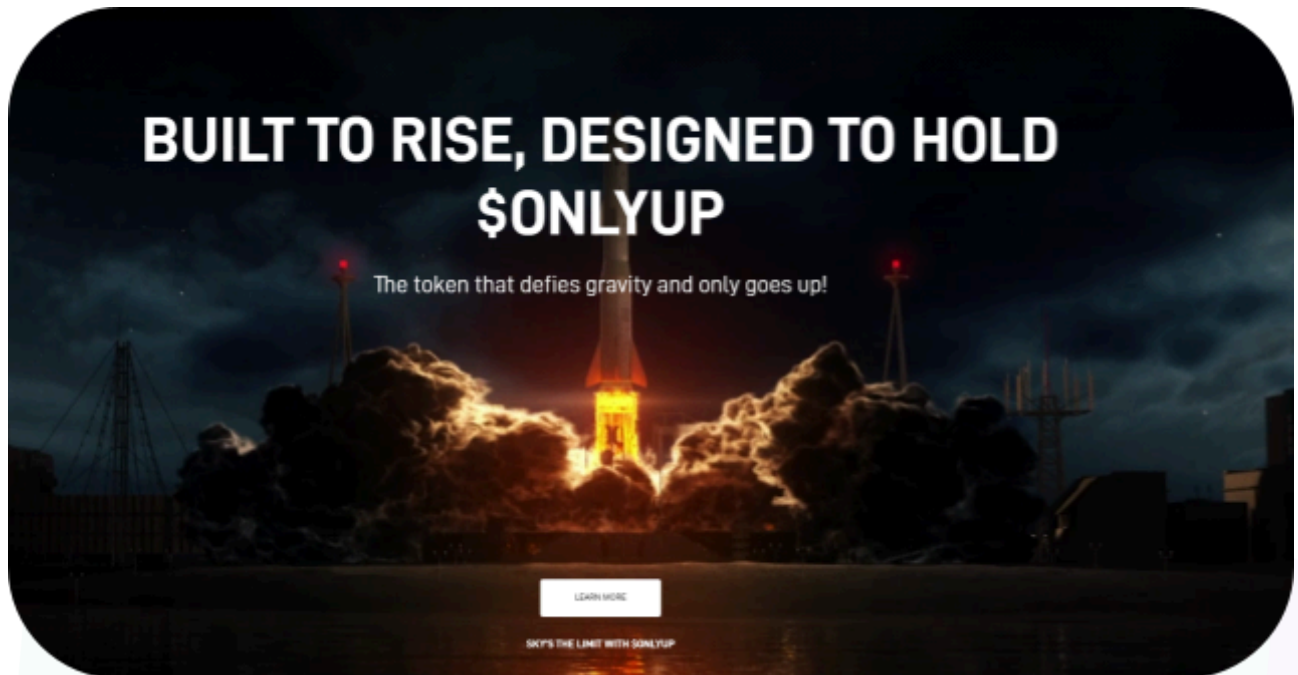
Website: www.onlyup.win

Logo:



Visualised Context:

Project Visuals:



Audit scope

We at Hashlock audited the solidity code within the OnlyUp project, the scope of work included a comprehensive review of the smart contracts listed below. We tested the smart contracts to check for their security, efficiency, and functionality. These tests were undertaken primarily through manual line-by-line analysis and were supported by software-assisted testing.

Description	OnlyUp Smart Contracts
Platform	EVM / Solidity
Audit Date	October, 2024
Contract 1	OnlyUpOnEthereum.sol
Contract 1 MD5 Hash	248ff818e29a511387ece5acaf9d676d
Contract 2	OnlyUpGeneric.sol
Contract 2 MD5 Hash	61b51c158bbb89874b913bfd6bfd1ffa
Fix Review Commit Hash	9a1d370f87e5fc145d5cec72a6821e8b0344e6e6

Security Rating

After Hashlock's Audit, we found the smart contracts to be **"Secure"**. The contracts all follow simple logic, with correct and detailed ordering. They use a series of interfaces, and the protocol uses a list of Open Zeppelin contracts.



Not Secure

Vulnerable

Secure

Hashlocked

The 'Hashlocked' rating is reserved for projects that ensure ongoing security via bug bounty programs or on chain monitoring technology.

All issues uncovered during automated and manual analysis were meticulously reviewed and applicable findings are presented in the [Audit Findings](#) section. The general security overview is presented in the [Standardised Checks](#) section and the project's contract functionality is presented in the [Intended Smart Contract Functions](#) section.

All findings we have identified have been resolved or acknowledged.

Hashlock found:

1 Low severity vulnerability

1 QA

Caution: *Hashlock's audits do not guarantee a project's success or ethics, and are not liable or responsible for security. Always conduct independent research about any project before interacting.*

Intended Smart Contract Functions

Claimed Behaviour	Actual Behaviour
OnlyUpOnEthereum.sol ERC20 token contract with an ordering functionality on the transfers	Contract achieves this functionality.
OnlyUpGeneric.sol Abstract for the OnlyUpOnEthereum contract	Contract achieves this functionality.

Code Quality

This audit scope involves the smart contracts of the OnlyUp project, as outlined in the Audit Scope section. All contracts, libraries, and interfaces mostly follow standard best practices and to help avoid unnecessary complexity that increases the likelihood of exploitation, however, some refactoring was required.

The code is very well commented on and closely follows best practice nat-spec styling. All comments are correctly aligned with code functionality.

Audit Resources

We were given the OnlyUp project smart contract code in the form of Github access.

As mentioned above, code parts are well commented. The logic is straightforward, and therefore it is easy to quickly comprehend the programming flow as well as the complex code logic. The comments are helpful in providing an understanding of the protocol's overall architecture.

Dependencies

As per our observation, the libraries used in this smart contracts infrastructure are based on well-known industry standard open source projects.

Severity Definitions

Significance	Description
High	High-severity vulnerabilities can result in loss of funds, asset loss, access denial, and other critical issues that will result in the direct loss of funds and control by the owners and community.
Medium	Medium-level difficulties should be solved before deployment, but won't result in loss of funds.
Low	Low-level vulnerabilities are areas that lack best practices that may cause small complications in the future.
Gas	Gas Optimisations, issues, and inefficiencies

Audit Findings

Low

[L-01] OnlyUpGeneric#SetBlacklist, SetEolDelta, SetAutoUnlockTimespan, EnableOpenMarket, WhitelistEarlyBuy - Lack of emitting events

Description

The SetBlacklist, SetEolDelta, SetAutoUnlockTimespan, EnableOpenMarket, and WhitelistEarlyBuy functions are missing events when key storages are updated.

Recommendation

Add relevant events in the functions.

Status

Resolved

QA

[Q-01] OnlyUpGeneric#LaunchPair - Lack of checking for a return value

Description

The LaunchPair function is missing a check of return value on the wEth transfer.

Recommendation

Add a check to make sure that the return value on the token transfer is true in the function.

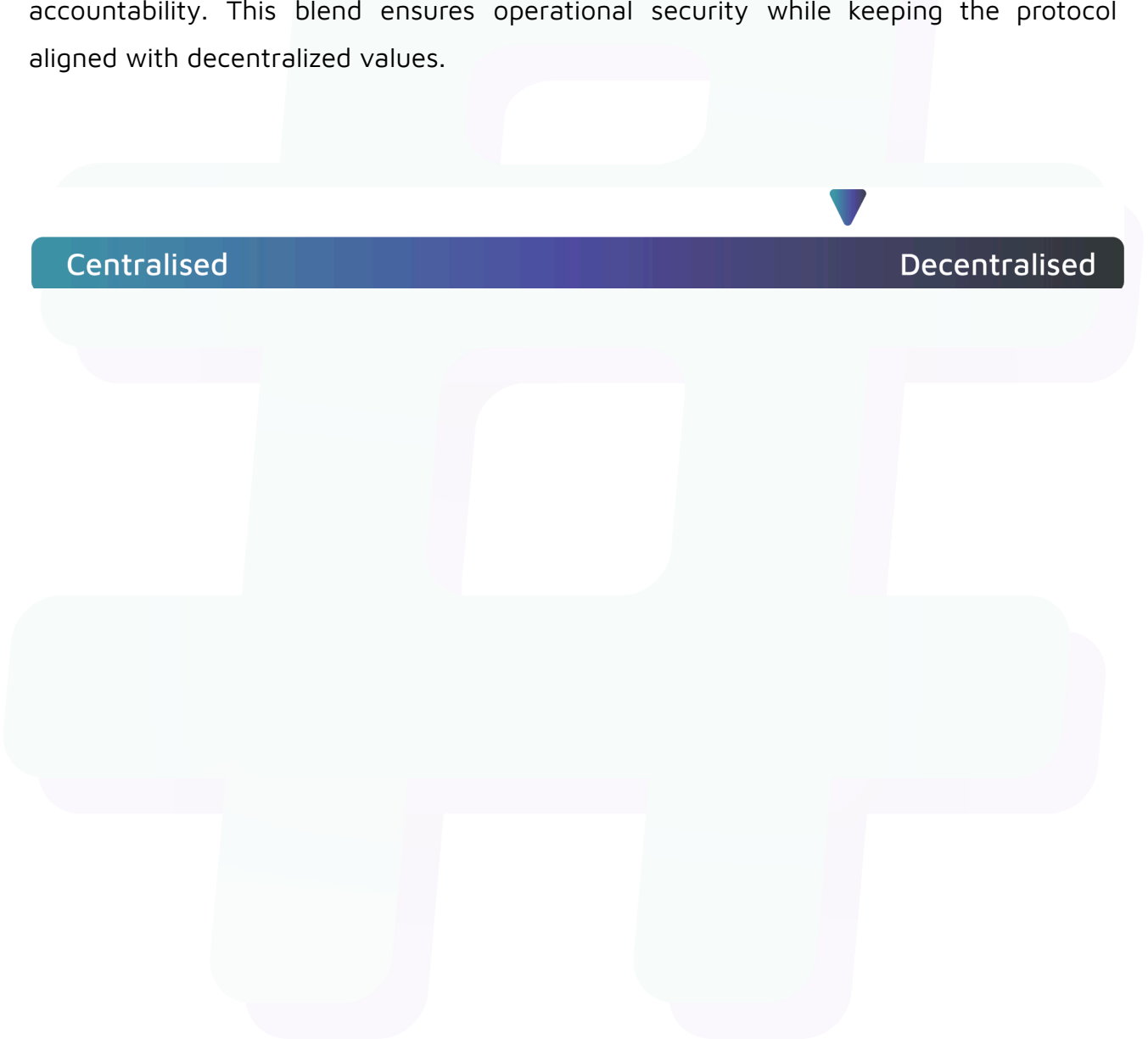
Status

Resolved

Centralisation

The OnlyUp project values community-centered and transparency over centralisation.

OnlyUp emphasizes a community-centric approach while maintaining transparency. Although some critical functions remain centralized for enhanced security and functionality, the project strives to balance decentralization with internal team accountability. This blend ensures operational security while keeping the protocol aligned with decentralized values.



Centralised

Decentralised

Conclusion

After Hashlocks analysis, the OnlyUp project seems to have a sound and well-tested code base, now that our vulnerability findings have been resolved and acknowledged. Overall, most of the code is correctly ordered and follows industry best practices. The code is well commented on as well. To the best of our ability, Hashlock is not able to identify any further vulnerabilities.

Our Methodology

Hashlock strives to maintain a transparent working process and to make our audits a collaborative effort. The objective of our security audits is to improve the quality of systems and upcoming projects we review and to aim for sufficient remediation to help protect users and project leaders. Below is the methodology we use in our security audit process.

Manual Code Review:

In manually analysing all of the code, we seek to find any potential issues with code logic, error handling, protocol and header parsing, cryptographic errors, and random number generators. We also watch for areas where more defensive programming could reduce the risk of future mistakes and speed up future audits. Although our primary focus is on the in-scope code, we examine dependency code and behaviour when it is relevant to a particular line of investigation.

Vulnerability Analysis:

Our methodologies include manual code analysis, user interface interaction, and white box penetration testing. We consider the project's website, specifications, and whitepaper (if available) to attain a high-level understanding of what functionality the smart contract under review contains. We then communicate with the developers and founders to gain insight into their vision for the project. We install and deploy the relevant software, exploring the user interactions and roles. While we do this, we brainstorm threat models and attack surfaces. We read design documentation, review other audit results, search for similar projects, examine source code dependencies, skim open issue tickets, and generally investigate details other than the implementation.

Documenting Results:

We undergo a robust, transparent process for analysing potential security vulnerabilities and seeing them through to successful remediation. When a potential issue is discovered, we immediately create an issue entry for it in this document, even though we have not yet verified the feasibility and impact of the issue. This process is vast because we document our suspicions early even if they are later shown to not represent exploitable vulnerabilities. We generally follow a process of first documenting the suspicion with unresolved questions, and then confirming the issue through code analysis, live experimentation, or automated tests. Code analysis is the most tentative, and we strive to provide test code, log captures, or screenshots demonstrating our confirmation. After this, we analyse the feasibility of an attack in a live system.

Suggested Solutions:

We search for immediate mitigations that live deployments can take and finally, we suggest the requirements for remediation engineering for future releases. The mitigation and remediation recommendations should be scrutinised by the developers and deployment engineers, and successful mitigation and remediation is an ongoing collaborative process after we deliver our report, and before the contract details are made public.

Disclaimers

Hashlock's Disclaimer

Hashlock's team has analysed these smart contracts in accordance with the best industry practices at the date of this report, in relation to: cybersecurity vulnerabilities and issues in the smart contract source code, the details of which are disclosed in this report, (Source Code); the Source Code compilation, deployment, and functionality (performing the intended functions).

Due to the fact that the total number of test cases is unlimited, the audit makes no statements or warranties on the security of the code. It also cannot be considered as a sufficient assessment regarding the utility and safety of the code, bug-free status, or any other statements of the contract. While we have done our best in conducting the analysis and producing this report, it is important to note that you should not rely on this report only. We also suggest conducting a bug bounty program to confirm the high level of security of this smart contract.

Hashlock is not responsible for the safety of any funds and is not in any way liable for the security of the project.

Technical Disclaimer

Smart contracts are deployed and executed on a blockchain platform. The platform, its programming language, and other software related to the smart contract can have their own vulnerabilities that can lead to attacks. Thus, the audit can't guarantee the explicit security of the audited smart contracts.

About Hashlock

Hashlock is an Australian-based company aiming to help facilitate the successful widespread adoption of distributed ledger technology. Our key services all have a focus on security, as well as projects that focus on streamlined adoption in the business sector.

Hashlock is excited to continue to grow its partnerships with developers and other web3-oriented companies to collaborate on secure innovation, helping businesses and decentralised entities alike.

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